



NORTH SYDNEY BOYS HIGH SCHOOL

2013 HSC ASSESSMENT TASK 2

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 55 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Lucas
- ☐ Mr Berry
- ☐ Mr Lin
- ☐ Mr Fletcher
- ☐ Mr Lam
- ☐ Ms Ziazaris

Student Name/Number: _____

(To be used by the exam markers only.)

Question No	1-5	6	7	8	9	10	11	Total	Total %
Mark	5	10	8	6	10	3	7	49	100

SECTION A – MULTIPLE CHOICE (5 Marks)

1. If $y = e^{x^2}$ which of the following is an expression for $\frac{d^2y}{dx^2}$
- (A) $2xe^{x^2}$
(B) $2e^{x^2}$
(C) $4xe^{x^2}$
(D) $(4x^2 + 2)e^{x^2}$
2. If $e^{x+2} = 4$, the exact value of x is:
- (A) 2
(B) $e^4 + 2$
(C) $\log_e 2$
(D) $\log_e 4 - 2$
3. The polynomial $P(x) = ax^3 + bx^2 + cx + d$ has zeros at: $-2, -1, 1$ and 2 . What is the value of b
- (A) -2
(B) -1
(C) 0
(D) 2
4. The function $y = x^3 + 2x^2 + 3x + 4$ has a root between -1 and -2 . By halving the interval a better approximation can be determined to be between:
- (A) -1 and -1.25
(B) -1.25 and -1.5
(C) -1.5 and -1.75
(D) -1.75 and -2
5. The gradient of the normal to the curve $y = \log_e x^2$ at $x = 2$ is:
- (A) -1
(B) 0
(C) 1
(D) 4

SECTION B – EXTENDED RESPONSE

Question 6 (10 Marks)

(a) Differentiate

(i) $\log_{10} x$ 1

(ii) $x^2 e^x$ 2

(b) Find the primitive of:

(i) $2e^x + 1$ 1

(ii) $\frac{x}{2x^2 + 1}$ 2

(iii) $\frac{2x^2 + 1}{x}$ 2

(c) Evaluate $\int_1^3 e^{2x} dx$ correct to two decimal places: 2

Question 7 (8 Marks)

(a) (i) Factorise $x^3 - x^2 - 8x + 12$ completely 3

(ii) Hence sketch $P(x) = x^3 - x^2 - 8x + 12$ 2
(you are not required to use calculus)

(b) The root of $f(x) = 0.4x - e^{-x^2}$ is near $x = 1$. Use Newton's Method once to find a 2 decimal place approximation to this root 3

Question 8 (6 Marks)

(a) If α, β and γ are the roots of the equation $3x^3 - 5x^2 + 2x - 3 = 0$.
Find the values of

(i) $\alpha + \beta + \gamma$ 1

(ii) $\alpha\beta + \beta\gamma + \alpha\gamma$ 1

(iii) $\alpha^2 + \beta^2 + \gamma^2$ 1

(b) The polynomial $P(x) = x^3 + ax^2 - bx + 6$ has $(x - 3)$ as a factor.
When $P(x)$ is divided by $(x + 1)$ the remainder is 8.
Find the values of a and b . 3

Question 9 (10 Marks)

- (a) Find the volume when $y = \frac{2}{\sqrt{x}}$ is rotated around the x -axis between $x = 3$ and $x = 5$ **3**
- (b) (i) Draw a neat sketch of $y = \log_e(x - 1)$ showing all relevant features **2**
- (ii) Use 2 applications (i.e. five function values) of Simpson's rule to estimate the area enclosed by $y = \log_e(x - 1)$, $x = 6$ and the x -axis, in the first quadrant correct to two decimal places. **3**
- (iii) Hence, or otherwise, approximate the area enclosed by the lines $y = \log_e(x - 1)$, $y = \log_e 5$ and the coordinate axes. **2**

Question 10 (3 Marks)

- (a) Prove by mathematical induction that $7^n - 6n - 1$ is divisible by 36 for all positive integers $n \geq 2$. **3**

Question 11 (7 Marks)

- (a) Solve for x : $\log_2 x + \log_2(x + 7) = 3$ **3**
- (b) If $y = \frac{\log_e x}{x}$
- (i) Show that $\frac{dy}{dx} = \frac{1 - \log_e x}{x^2}$ **1**
- (ii) Hence or otherwise show that:
- $$\int_e^{e^2} \frac{1 - \log_e x}{x \log_e x} dx = (\log_e 2) - 1$$
- 3**

End of Examination

MULTIPLE CHOICE

1. D

2. D

3. C

4. C

5. A

Question 6

$$(a)(i) \frac{d}{dx} \log_{10} x = \frac{1}{x \log_e 10}$$

✓ correct answer

$$(a)(ii) y = x^2 e^x$$

$$\text{let } u = x^2 \\ \therefore u' = 2x$$

$$\text{let } v = e^x \\ \therefore v' = e^x$$

$$y' = u'v + v'u$$

$$= (2x)(e^x) + (e^x)(x^2)$$

$$= x e^x (x + 2)$$

✓ product rule

✓ answer

$$(b)(i) \int (2e^x + 1) dx = 2e^x + x + C$$

✓ correct answer

$$(b)(ii) \int \frac{x}{2x^2 + 1} dx$$

$$= \frac{1}{4} \int \left(\frac{4x}{2x^2 + 1} \right) dx$$

$$= \frac{1}{4} \log_e (2x^2 + 1) + C$$

✓ any log as the primitive

✓ correct answer

$$(b)(iii) \int \frac{2x^2 + 1}{x} dx$$

$$= \int \left(2x + \frac{1}{x} \right) dx$$

$$= x^2 + \log_e x + C$$

✓ breaking up fraction

✓ correct answer

$$(c) \int_1^3 e^{2x} dx$$

$$= \left[\frac{1}{2} e^{2x} \right]_1^3$$

$$= \frac{1}{2} (e^6 - e^2)$$

$$= 198.02 \text{ (2 d.p.)}$$

✓ correct primitive

✓ correct answer

Question 7

(a)(i) let $P(x) = x^3 - x^2 - 8x + 12$

$$P(2) = 2^3 - 2^2 - 8 \times 2 + 12$$

$$= 0$$

$\therefore (x-2)$ is a factor

✓ finding a factor

$$\begin{array}{r} x^2 + x - 6 \\ x-2 \overline{) x^3 - x^2 - 8x + 12} \\ \underline{x^3 - 2x^2} \\ x^2 - 8x \\ \underline{x^2 - 2x} \\ -6x + 12 \\ \underline{-6x + 12} \\ 0 \end{array}$$

✓ polynomial division

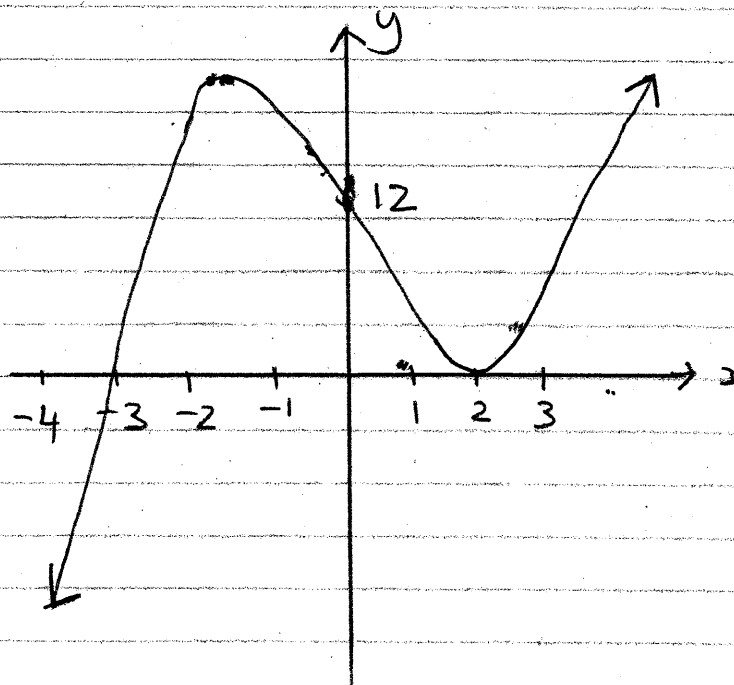
$\therefore P(x) = (x-2)(x^2 + x - 6)$

$\therefore P(x) = (x-2)(x+3)(x-2)$

✓ factorised expression

$$P(x) = (x-2)^2(x+3)$$

(a)(ii)



✓ correct single & double root

✓ shape

$$(b) f(x) = 0.4x - e^{-x^2}$$

$$\therefore f'(x) = 0.4 + 2xe^{-x^2}$$

✓ correct derivative

$$a_0 = 1$$

$$a_1 = a_0 - \frac{f(a_0)}{f'(a_0)}$$

$$= 1 - \frac{0.4 - e^{-1}}{0.4 + 2e^{-1}}$$

✓ subbing $a_0 = 1$ into formula

$$\therefore a_1 = 0.97 \text{ (2 d.p.)}$$

✓ final answer

Question 8

$$\begin{aligned} \text{(a)(i)} \quad \alpha + \beta + \gamma &= -b/a \\ &= 5/3 \end{aligned}$$

✓ correct answer

$$\begin{aligned} \text{(a)(ii)} \quad \alpha\beta + \alpha\gamma + \beta\gamma &= c/a \\ &= 2/3 \end{aligned}$$

✓ correct answer

$$\begin{aligned} \text{(a)(iii)} \quad \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= (5/3)^2 - 2(2/3) \\ &= 13/9 \end{aligned}$$

✓ correct answer

$$\text{(b)} \quad P(3) = 0$$

$$\therefore 3^3 + a \times 3^2 - 3b + 6 = 0$$

$$33 + 9a - 3b = 0$$

$$9a - 3b = -33 \dots \textcircled{1}$$

✓ simultaneous equations

$$P(-1) = 8$$

$$\therefore (-1)^3 + a(-1)^2 + b + 6 = 8$$

$$-1 + a + b + 6 = 8$$

$$a + b = 3$$

$$\therefore b = 3 - a \dots \textcircled{2}$$

$$\text{sub } \textcircled{2} \rightarrow \textcircled{1}$$

$$9a - 3(3 - a) = -33$$

$$9a - 9 + 3a = -33$$

$$12a = -24$$

$$a = -2$$

✓ solution to a

sub into ②

$$b = 3 - a$$

$$= 3 - (-2)$$

$$\therefore b = 5$$

✓ solution to b

Question 9

$$(a) \quad V = \pi \int_a^b y^2 dx$$

$$= \pi \int_3^5 \left(\frac{2}{\sqrt{x}}\right)^2 dx$$

$$= \pi \int_3^5 4/x dx$$

$$= 4\pi \int_3^5 1/x dx$$

$$= 4\pi [\log_e x]_3^5$$

$$= 4\pi [\log_e 5 - \log_e 3]$$

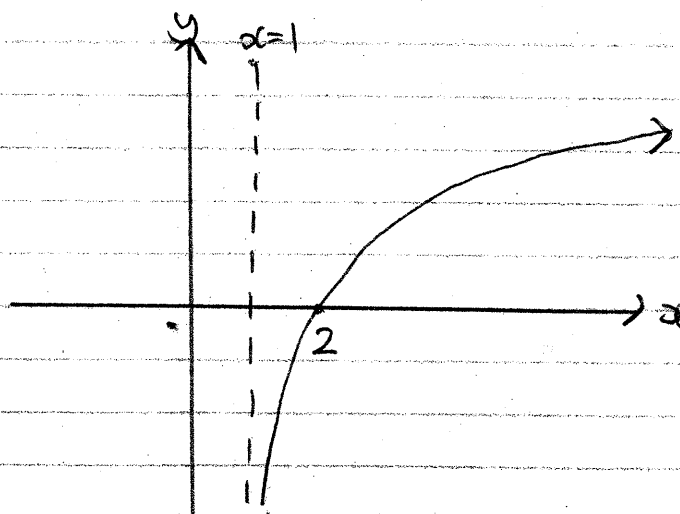
$$= 4\pi \log_e(5/3)$$

✓ subbing into formula

✓ correct integral

✓ final answer

(b)(i)



✓ shape

✓ asymptote AND x-intercept

(b)(ii)

y_0	y_1	y_2	y_3	y_4
$\log 1$	$\log 2$	$\log 3$	$\log 4$	$\log 5$

$$A \doteq \frac{1}{3} [(y_0 + y_n) + 4(y_1 + y_3 + \dots) + 2(y_2 + y_4 + \dots)]$$

$$= \frac{1}{3} [(\log 1 + \log 5) + 4(\log 2 + \log 4) + 2(\log 3)]$$

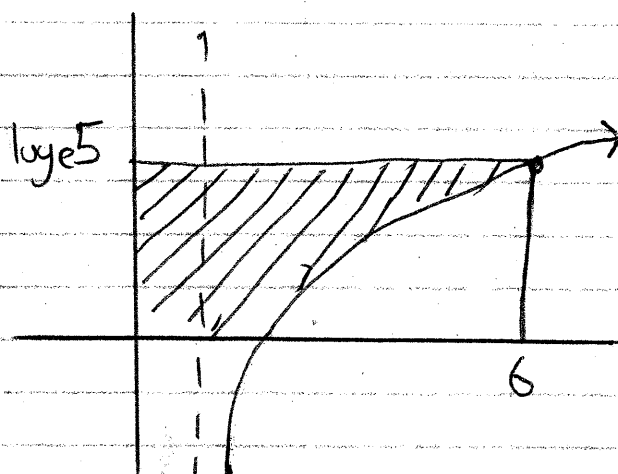
$$= 4.04 \text{ (2 d.p.)}$$

✓ correct formula

✓ correct values

✓ final answer

(b)(iii)



✓ area of
rectangle

✓ final
answer

(any rounding
is ok)

$$\begin{aligned} A &\doteq (6 \times \log_e 5) - 4.04 \\ &= 5.615 \text{ (3 d.p.)} \end{aligned}$$

Question 10

1./ when $n = 2$

$$7^2 - 6 \times 2 - 1 = 36$$

which is divisible by 36

✓ proving
for base
case

2./ assume true for $n = k$
where k is an integer

$$\text{i.e. } 7^k - 6k - 1 = 36m$$

where m is an integer

3./ for $n = k + 1$

$$7^{k+1} - 6(k+1) - 1$$

$$= 7 \cdot 7^k - 6k - 6 - 1$$

$$= 7 \cdot 7^k - 6k - 7$$

$$= 7 \times 7^k - 42k - 7 - 6k + 42k$$

$$= 7(7^k - 6k - 1) + 36k$$

$$= 7 \times 36m + 36k, \text{ from assumption}$$

$$= 36(7m + k)$$

✓ substitution
from assumption

which is divisible by 36 since
 m & k are both integers

✓ correctly
showing
divisibility

4./ therefore the proposition is
true by the principle of
mathematical induction

Question 11

$$(a) \log_2 x + \log_2(x+7) = 3$$

$$\log_2(x^2 + 7x) = 3$$

$$\therefore x^2 + 7x = 2^3$$

$$x^2 + 7x = 8$$

$$x^2 + 7x - 8 = 0$$

$$(x+8)(x-1) = 0$$

$$\therefore x = -8 \text{ or } x = 1$$

$$\therefore x = 1 \text{ as } x > 0$$

✓ combining into 1 log

✓ re-arranging to get exponential

✓ correct final answer must have only 1 final solution

$$(b)(i) y = \frac{\log_e x}{x}$$

$$\text{let } u = \log_e x \text{ \& } v = x$$

$$\therefore u' = \frac{1}{x} \text{ \& } v' = 1$$

$$\frac{dy}{dx} = \frac{u'v - v'u}{v^2}$$

$$= \frac{(\frac{1}{x})(x) - (1)\log_e x}{x^2}$$

$$= \frac{1 - \log_e x}{x^2} \text{ as req'd}$$

✓ all necessary steps shown

$$(b)(i) \int_e^{e^2} \frac{1 - \log_e x}{x \log_e x} dx$$

$$= \int_e^{e^2} \frac{\left(\frac{1 - \log_e x}{x^2} \right)}{\left(\frac{x \log_e x}{x^2} \right)} dx$$

✓ dividing by x^2

$$= \int_e^{e^2} \frac{\left(\frac{1 - \log_e x}{x^2} \right)}{\left(\frac{\log_e x}{x} \right)} dx$$

$$= \left[\log_e \left(\frac{\log_e x}{x} \right) \right]_e^{e^2} \quad (\text{from (i)})$$

✓ correct integral

$$= \log_e \left(\frac{\log_e e^2}{e^2} \right) - \log_e \left(\frac{\log_e e}{e} \right)$$

$$= \log_e \left(\frac{2}{e^2} \right) - \log_e \left(\frac{1}{e} \right)$$

$$= \log_e 2 - \log_e e^2 - \log_e 1 + \log_e e$$

$$= \log_e 2 - 2 + 0 + 1$$

$$= (\log_e 2) - 1, \text{ as required}$$

✓ successfully getting to final answer.